

REPORT

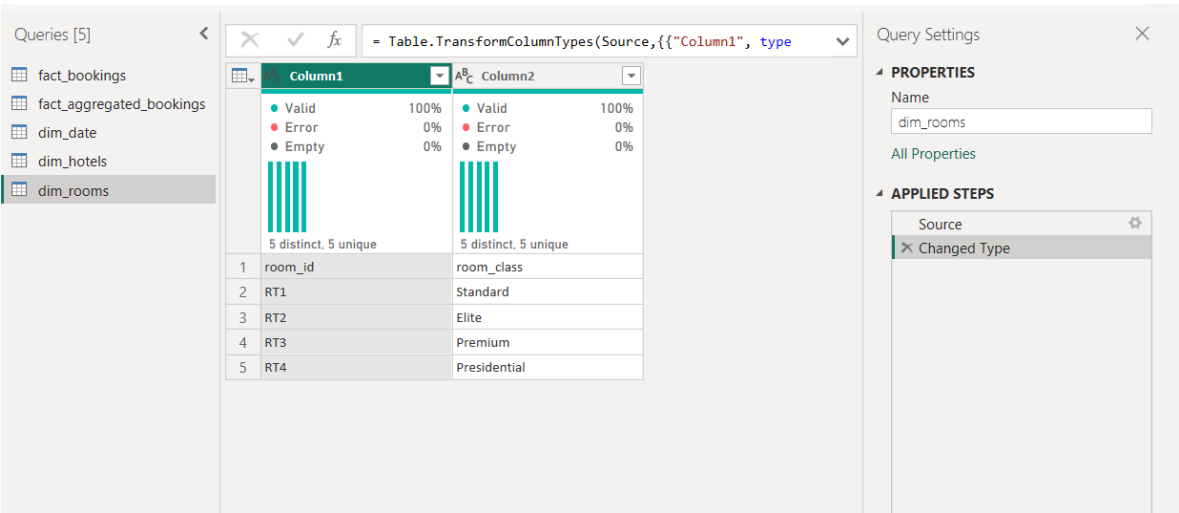
**Objective:** This project sets out to take the raw, multi-source hospitality dataset of AtliQ Grands a fictional five-star hotel chain in India, provided by Codebasics as part of their Resume Project Challenge and transform it into an interactive, insight-driven Power BI dashboard. The goal is to enable management to make data-driven decisions on revenue, occupancy, cancellations, and property performance to reverse the brand's declining market share and revenue.

**Tools & Libraries:** Power BI Desktop, Power Query (M language), DAX (Data Analysis Expressions)

**Dataset:** "Hospitality Revenue Insights" dataset from Codebasics — 5 relational tables:

- dim\_date — Date dimension (92 days, May–July 2022)
- dim\_hotels — 7 properties across 4 cities
- dim\_rooms — 4 room classes (Elite, Premium, Standard, Presidential)
- fact\_bookings — ~135,000 booking-level transactions
- fact\_aggregated\_bookings — Daily aggregated room capacity and bookings

**Preparation:** Loaded 5 CSVs into Power BI via **Get Data → Text/CSV → Transform Data**, ensuring full control over data types and cleanliness before loading into the data model.



MAIN WORK

| Step           | Action Taken                                                                             | Key Result / Output                                                                                            |
|----------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1. Discovery   | Loaded all 5 CSVs and inspected schema, data types, and column profiling in Power Query. | Identified 5 tables, ~135K bookings, 92-day date range (May–July 2022). Confirmed 0% error across all columns. |
| 2. Structuring | Standardized column data types: dates as Date, IDs as Text, revenue as Decimal,          | All tables ready for relational modeling; consistent keys.                                                     |

|                                        |                                                                                                                                                                                                     |                                                                                  |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                                        | counts as Whole Number. Ensured property_id matched across all tables.                                                                                                                              |                                                                                  |
| <b>3. Cleaning (dim_date)</b>          | Converted corrupted mmm yy column to clean Text ("May 22"). Confirmed 92 unique dates with 0% empty values.                                                                                         | Clean time dimension ready for time intelligence.                                |
| <b>4. Cleaning (dim_rooms)</b>         | Applied <b>Use First Row as Headers</b> to promote row 1 into column headers (room_id, room_class).                                                                                                 | Room dimension correctly structured.                                             |
| <b>5. Data Modeling</b>                | Built a <b>star schema</b> with 6 relationships — fact_bookings and fact_aggregated_bookings at the center, surrounded by dim_date, dim_hotels, and dim_rooms. Marked dim_date as a Date Table.     | Fully connected relational model enabling cross-filtering and time intelligence. |
| <b>6. DAX Measures (Base)</b>          | Created foundational measures: Total Bookings, Total Capacity, Total Cancelled, Total Checked Out, Total No Show, Total Successful Booking, Daily Booked/Sellable/Utilized Room Nights, No of Days. | 10 base measures forming the calculation backbone.                               |
| <b>7. DAX Measures (KPI)</b>           | Built business KPIs: ADR, Average Rate, Occupancy %, Realisation %, RevPAR, Cancellation %, No Show Rate %, Booking % by Platform, Booking % by Room Class.                                         | 9 KPI measures representing the hotel industry standard metric set.              |
| <b>8. DAX Measures (WoW)</b>           | Created 5 Week-over-Week change measures using CALCULATE + DATEADD(-7, DAY): ADR WoW %, RevPAR WoW %, Occupancy WoW %, Realisation WoW %, DSRN WoW %.                                               | 5 time-intelligence measures enabling trend analysis.                            |
| <b>9. Visualization</b>                | Designed a 4-page interactive dashboard with synced slicers (City, Room Class, Hotel Category, Month).                                                                                              | Full user-driven analytical experience.                                          |
| <b>10. Validation &amp; Publishing</b> | Verified measure totals (Cancelled + Checked Out + No Show = Total Bookings). Exported .pbix, screenshots, and pushed to GitHub.                                                                    | Ready for portfolio                                                              |

## THE 24 METRICS

### 1. Average daily rate

```

1 Average Daily Rate =
2 DIVIDE(
3     SUM(fact_bookings[revenue_realized]),
4     [Daily Utilized Room Nights]
5 )

```

## 2. week-over-week percentage change in Average daily rate

```

1 ADR WoW % =
2 VAR CurrentADR = [Average Daily Rate]
3 VAR PrevADR =
4     CALCULATE(
5         [Average Daily Rate],
6         DATEADD(dim_date[date], -7, DAY)
7     )
8 RETURN
9 DIVIDE(CurrentADR - PrevADR, PrevADR)

```

## 3. Average Rate

```

1 Average Rate =
2 DIVIDE(
3     SUM(fact_bookings[revenue_realized]),
4     [Total Successful Booking]
5 )

```

## 4. Booking % by platform

```

1 Booking % by Platform =
2 DIVIDE(
3     [Total Bookings],
4     CALCULATE(
5         [Total Bookings],
6         ALL(fact_bookings[booking_platform])
7     )
8 )

```

## 5. Booking % by room class

```

1 Booking % by Room Class =
2 DIVIDE(
3     [Total Bookings],
4     CALCULATE(
5         [Total Bookings],
6         ALL(dim_rooms[room_class])
7     )
8 )

```

## 6. cancellation %

```

1 Cancellation % =
2 DIVIDE(
3     [Total Cancelled Bookings],
4     [Total Bookings]
5 )

```

## 7. Daily Booked Room Night

```
1 Daily Booked Room Nights = SUM(fact_aggregated_bookings[successful_bookings])
```

## 8. Daily Sellable Room Nights

```
1 Daily Sellable Room Nights = SUM(fact_aggregated_bookings[capacity])
```

## 9. Daily Sellable Room Nights — Week-over-Week change

```
1 DSRN WoW % =  
2 VAR CurrentDSRN = [Daily Sellable Room Nights]  
3 VAR PrevDSRN =  
4     CALCULATE(  
5         [Daily Sellable Room Nights],  
6         DATEADD(dim_date[date], -7, DAY)  
7     )  
8 RETURN  
9 DIVIDE(CurrentDSRN - PrevDSRN, PrevDSRN)
```

## 10. Daily Utilized Room Nights

```
1 Daily Utilized Room Nights = [Total Checked Out]
```

## 11. No of Days

```
1 No of Days = DISTINCTCOUNT(dim_date[date])
```

## 12. No show rate %

```
1 No Show Rate % =  
2 DIVIDE(  
3     [Total No Show Booking],  
4     [Total Bookings]  
5 )
```

## 13. Occupancy%

```
1 Occupancy % =  
2 DIVIDE(  
3     [Daily Booked Room Nights],  
4     [Daily Sellable Room Nights]  
5 )
```

## 14. Occupancy Week over Week change

```
1 Occupancy WoW % =  
2 VAR CurrentOcc = [Occupancy %]  
3 VAR PrevOcc =  
4     CALCULATE(  
5         [Occupancy %],  
6         DATEADD(dim_date[date], -7, DAY)  
7     )  
8 RETURN  
9 DIVIDE(CurrentOcc - PrevOcc, PrevOcc)
```

## 15. Realisation%

```
1 Realisation % =  
2 DIVIDE(  
3     SUM(fact_bookings[revenue_realized]),  
4     SUM(fact_bookings[revenue_generated])  
5 )
```

## 16. Realisation Week over week change

```
1 Realisation WoW % =  
2 VAR CurrentReal = [Realisation %]  
3 VAR PrevReal =  
4     CALCULATE(  
5         [Realisation %],  
6         DATEADD(dim_date[date], -7, DAY)  
7     )  
8 RETURN  
9 DIVIDE((CurrentReal - PrevReal), PrevReal)
```

## 17. Revenue per Available Room

```
1 Revenue per Available Room =  
2 DIVIDE(  
3     SUM(fact_bookings[revenue_realized]),  
4     [Total Capacity]  
5 )
```

## 18. Revenue per Available Room week over week change

```
1 RevPAR WoW % =  
2 VAR CurrentRevPAR = [Revenue per Available Room]  
3 VAR PrevRevPAR =  
4     CALCULATE(  
5         [Revenue per Available Room],  
6         DATEADD(dim_date[date], -7, DAY)  
7     )  
8 RETURN  
9 DIVIDE((CurrentRevPAR - PrevRevPAR), PrevRevPAR)
```

## 19. Total bookings

```
1 Total Bookings = COUNTROWS(fact_bookings)
```

## 20. Total Cancelled bookings

```
1 Total Cancelled Bookings =  
2 CALCULATE(  
3     [Total Bookings],  
4     fact_bookings[booking_status] = "Cancelled"  
5 )
```

## 21. Total Capacity

```
1 Total Capacity = SUM(fact_aggregated_bookings[capacity])
```

## 22. Total Checked Out

```
1 Total Checked Out =  
2 CALCULATE(  
3     [Total Bookings],  
4     fact_bookings[booking_status] = "Checked Out"  
5 )
```

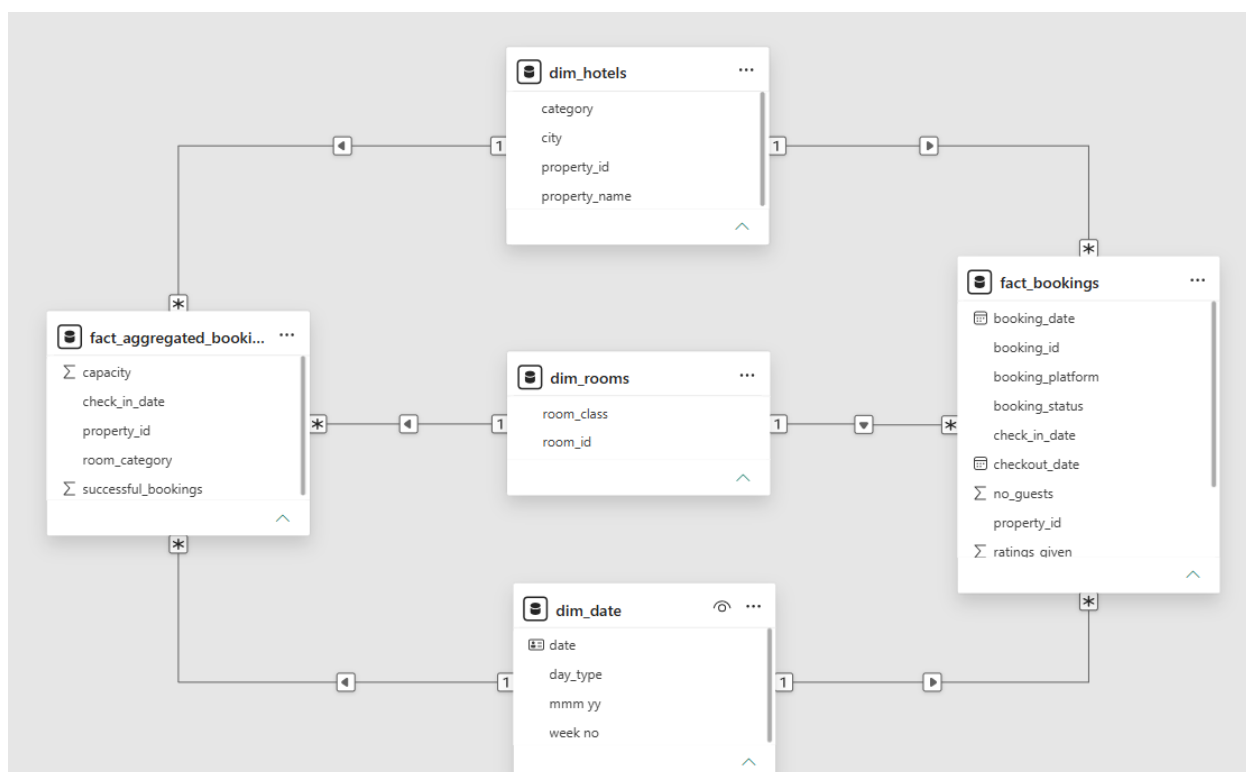
## 23. Total No show booking

```
1 Total No Show Booking =  
2 CALCULATE(  
3     [Total Bookings],  
4     fact_bookings[booking_status] = "No Show"  
5 )  
6
```

## 24. Total Successful Booking

```
1 Total Successful Booking =  
2 CALCULATE(  
3     [Total Bookings],  
4     fact_bookings[booking_status] IN {"Checked Out", "No Show"}  
5 )
```

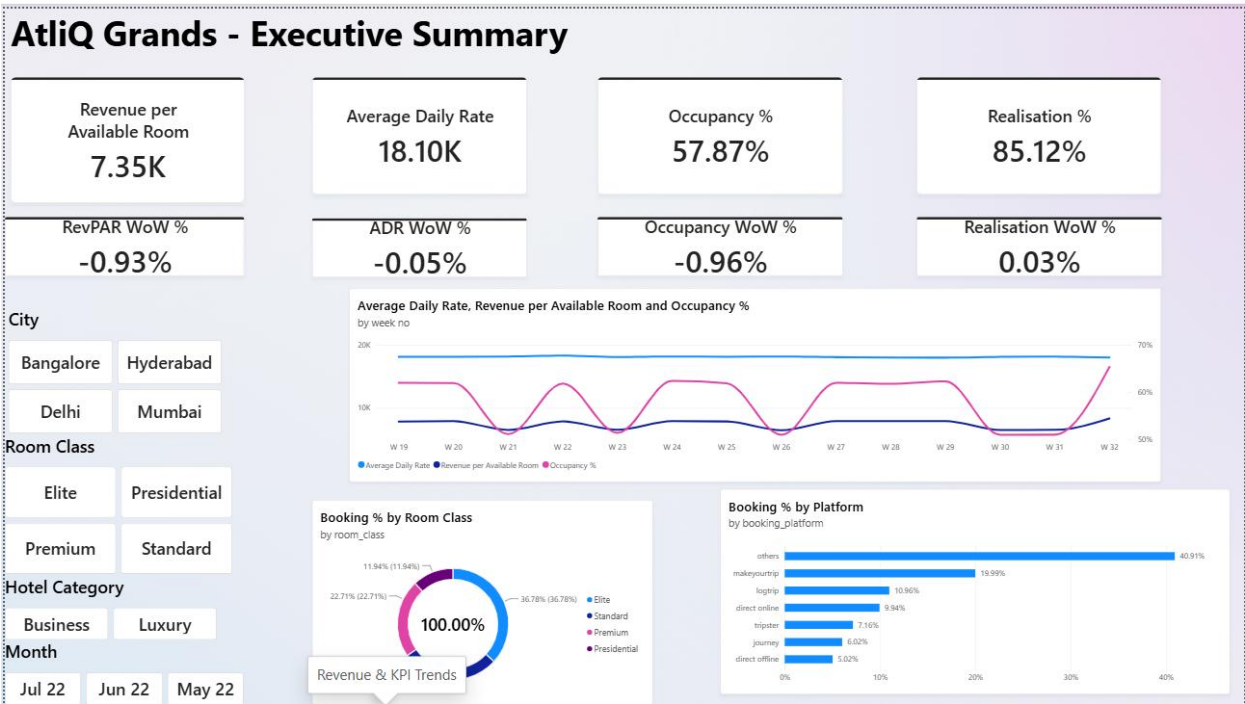
## STAR SCHEMA



## 5. DASHBOARD PAGES

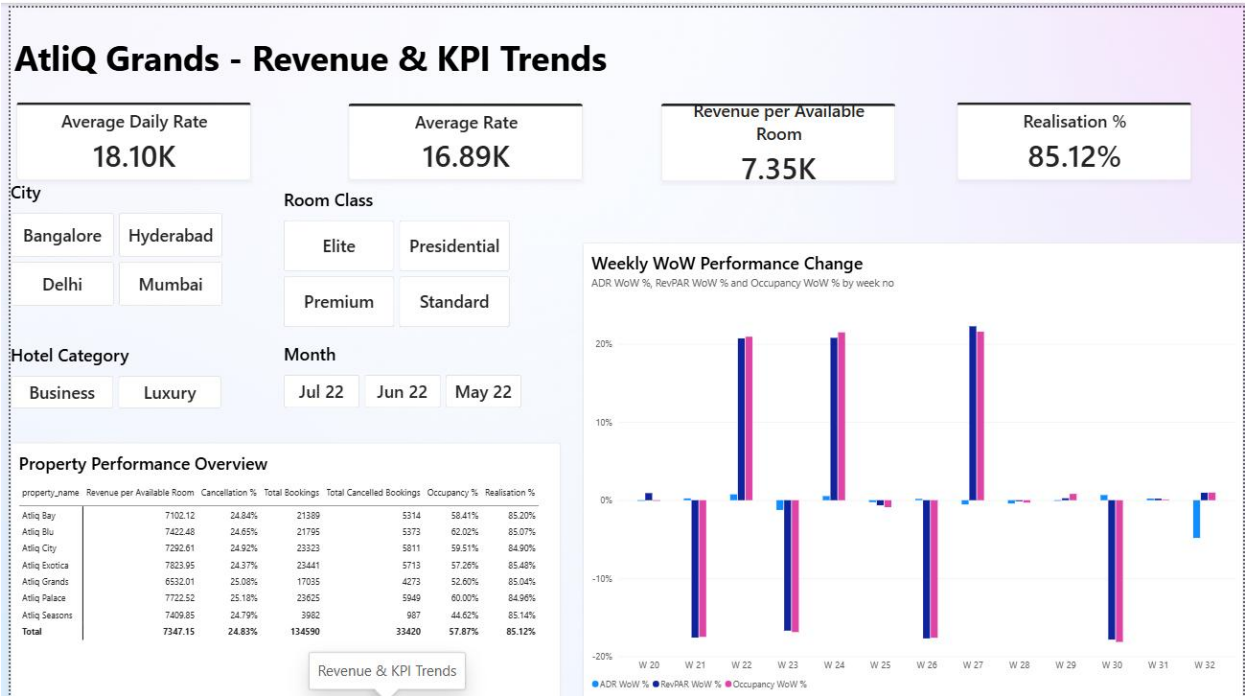
Page 1: Home (Executive Summary)

High-level snapshot for leadership — 4 main KPIs (RevPAR, ADR, Occupancy %, Realisation %) with WoW changes, weekly trend line, and booking distribution by room class and platform.



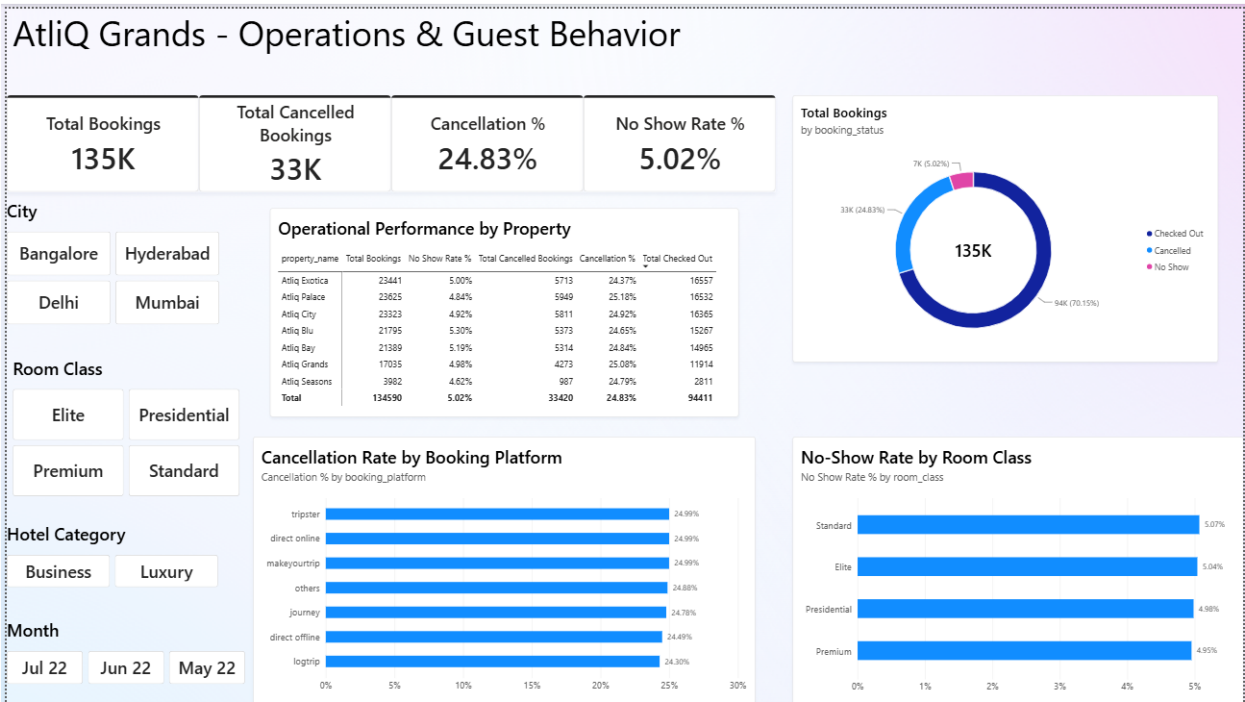
Page 2: Revenue & KPI Trends

Deeper dive into revenue drivers — 4 financial KPIs, a 3-metric WoW comparison chart, and a property-level performance table.



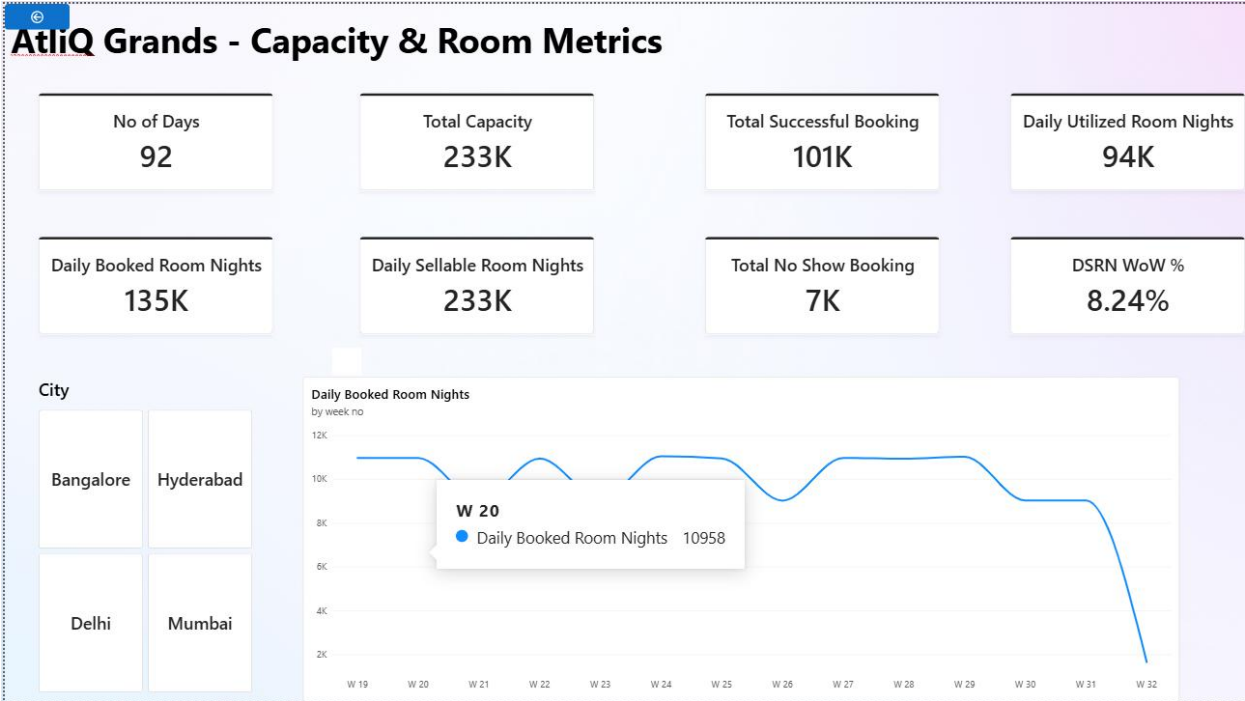
Page 3: Operations & Guest Behavior

Cancellation and no-show analysis by platform and room class, plus a booking status donut and property operational scorecard.



Page 4: Capacity & Room Metrics

Focused page covering remaining operational measures capacity, room nights (booked/sellable/utilized), successful bookings, no-shows, and a weekly booking trend line.



Key Metrics & Outputs:

- Total Bookings: 135,000 across 7 properties in 4 cities over 92 days

- **Total Revenue Realized:** ~₹1.7 billion (aggregate)
- **Average Occupancy:** 57.87% — healthy but with clear headroom
- **Average Daily Rate (ADR):** ₹18,100
- **RevPAR:** ₹7,350
- **Realisation %:** 85.12% — ~15% revenue lost to cancellations/no-shows
- **Cancellation Rate:** 24.83% — systemic across all platforms
- **No-Show Rate:** 5.02% — low and manageable

#### Validation Confirmation:

Measure totals reconciled — Cancelled (33K) + Checked Out (94K) + No Show (7K) = Total Bookings (135K).

#### Key Insights Uncovered:

1. Cancellation rate is high (~25%) and consistent across all booking platforms which is a systemic issue, not channel-specific.
2. Weekend occupancy clearly peaks, creating predictable demand spikes suitable for dynamic pricing.
3. RevPAR tracks occupancy closely, indicating occupancy is the primary revenue driver at current pricing.
4. ADR remains stable (~₹18K) with minimal week-to-week variation — pricing power is untested.
5. Certain properties (e.g., Atliq Seasons) contribute significantly less volume — possible expansion opportunity.

**Summary:** The project successfully transformed raw hospitality data into a 4-page interactive Power BI dashboard supported by a clean star schema and 24 purpose-built DAX measures. Through a disciplined pipeline of Load → Transform → Model → Measure → Visualize, the final deliverable gives management clear visibility into revenue drivers, operational weak points, and weekly performance trends.